



12586 - Pioneering Supernova Dust Studies with JWST: Detection of Dust Precursors in SN 2024ahv

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Phase 7				
	7	NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(1) 2024ahv
	8	MIRI/LRS	MIRI Low Resolution Spectroscopy	(1) 2024ahv
	9	MIRI/Imaging	MIRI Imaging	(1) 2024ahv
Phase 8				
	10	NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(1) 2024ahv
	11	MIRI/LRS	MIRI Low Resolution Spectroscopy	(1) 2024ahv
	12	MIRI/Imaging	MIRI Imaging	(1) 2024ahv

ABSTRACT

The rarity of nearby stripped-envelope supernovae (SESNe) presents an exceptional opportunity to study dust formation and evolution in the early Universe. SN 2024ahv, a rare Type Ib SESN (SN Ib) just 20 Mpc away, presents an unparalleled opportunity to study dust formation and evolution. As the closest SN Ib likely to explode during JWST's lifespan, it offers a unique chance to build a multi-wavelength legacy dataset, providing insights into massive star deaths and cosmic dust formation. JWST has already delivered remarkable early detections of CO and SiO in SN 2024ahv, making this supernova a critical probe for dust formation processes. While dust production in hydrogen-rich SNe II has been the primary focus until now, SESNe, born from short-lived, massive stars, are likely key contributors to early dust in the cosmos. Monitoring SN 2024ahv across NIR and MIR wavelengths will allow us to trace molecular evolution, dust formation, and composition in exquisite detail, offering insights never before captured for an SESN.

OBSERVING DESCRIPTION

We request 27.2 hours of charged time across four epochs over Cycles 4 and 5 (13.4 hrs in Cycle 4 and 13.8 hrs in Cycle 5) to obtain NIR+MIR time series of SN2024ahv between 500–1100 days. These observations will track molecular evolution and dust formation at key stages, with spectra at approximately 550, 800, 950, and 1100 days to monitor CO and SiO, as well as distinguish between O-rich and C-rich dust features.

Proposal 12586 - Targets - Pioneering Supernova Dust Studies with JWST: Detection of Dust Precursors in SN 2024ahv

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	2024ahv	RA: 16 18 46.3190 (244.6929958d) Dec: +07 24 44.77 (7.41244d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0 Epoch of Position: 2000	
	Comments: Category=Star Description=[Supernovae] Extended=NO				
Fixed Targets	(2)	offet_star	RA: 16 18 47.2960 (244.6970667d) Dec: +07 24 51.23 (7.41423d) Equinox: J2000	Proper Motion RA: -18.28975789727942 mas/yr Proper Motion Dec: -17.89740252837372 mas/yr Parallax: 0.002368960" Epoch of Position: 2016.0	
	Comments: Category=Star Description=[Disk stars] Extended=NO				

Proposal 12586 - Observation 7 - Pioneering Supernova Dust Studies with JWST: Detection of Dust Precursors in SN 2024ahv

Observation	Proposal 12586, Observation 7: NIRSPEC											Tue Jun 09 18:01:23 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	2024ahv	RA: 16 18 46.3190 (244.6929958d) Dec: +07 24 44.77 (7.41244d) Equinox: J2000				Proper Motion RA: 0 Proper Motion Dec: 0 Epoch of Position: 2000					
	Comments: Category=Star Description=[Supernovae] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	2 offet_star	WATA	SUB32	F140X	NRSRAPID	3	1	1	0.08	227508.2	
Template	HFF Readout Mode				Slit				Subarray			
	false				S400A1				SUBS400A1			
Dithers	#	Primary Dither Positions							Sub-Pixel Pattern			
	1	3							NONE			
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235M/F170LP	S400A1	NRSRAPID	100	2	1	NONE	3	6	944.271	227508.4
	2	G395M/F290LP	S400A1	NRSRAPID	100	2	2	NONE	3	6	944.271	227508.6

Proposal 12586 - Observation 7 - Pioneering Supernova Dust Studies with JWST: Detection of Dust Precursors in SN 2024ahv

Special Requirements	Between Dates 01-JUL-2026:00:00:00 and 30-AUG-2026:00:00:00 Group Observations 7, 8, 9 within 1 Days
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Proposal 12586 - Observation 8 - Pioneering Supernova Dust Studies with JWST: Detection of Dust Precursors in SN 2024ahv

Observation	Proposal 12586, Observation 8: MIRI/LRS										Tue Jun 09 18:01:23 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: MIRI Low Resolution Spectroscopy										
	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	2024ahv	RA: 16 18 46.3190 (244.6929958d) Dec: +07 24 44.77 (7.41244d) Equinox: J2000			Proper Motion RA: 0 Proper Motion Dec: 0 Epoch of Position: 2000					
Acquisition	Comments: Category=Star Description=[Supernovae] Extended=NO										
	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID		
Template	1	2 offset_star	F560W	FAST	4	1	1	11.1	307927.1		
	Subarray					Obtain Verification Image?					
Dithers	FULL					true					
	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset	
Pointing Verification	1	ALONG SLIT NOD									
	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter	
	1	FASTR1	6	1	1	1	1	16.65		F1000W	

Proposal 12586 - Observation 8 - Pioneering Supernova Dust Studies with JWST: Detection of Dust Precursors in SN 2024ahv

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	100	8	16	1	2	4478.915	227508.3
Special Requirements	Between Dates 01-JUL-2026:00:00:00 and 30-AUG-2026:00:00:00								
	Group Observations 7, 8, 9 within 1 Days								

Proposal 12586 - Observation 9 - Pioneering Supernova Dust Studies with JWST: Detection of Dust Precursors in SN 2024ahv

Observation	Proposal 12586, Observation 9: MIRI/Imaging										Tue Jun 09 18:01:23 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	2024ahv	RA: 16 18 46.3190 (244.6929958d)			Proper Motion RA: 0					
			Dec: +07 24 44.77 (7.41244d)			Proper Motion Dec: 0					
			Equinox: J2000			Epoch of Position: 2000					
		Comments: Category=Star Description=[Supernovae] Extended=NO									
Template	Subarray										
	BRIGHTSKY										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				1	1	POINT SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F1280W	FASTR1	50	4	1	Dither 1	4	16	702.607	227508.7
	2	F1500W	FASTR1	50	4	1	Dither 1	4	16	702.607	227508.9
	3	F1800W	FASTR1	50	4	1	Dither 1	4	16	702.607	227508.10
	4	F2100W	FASTR1	70	8	1	Dither 1	4	32	1962.455	227508.11
	5	F2550W	FASTR1	50	13	1	Dither 1	4	52	2291.261	227508.12
Special Requirements	Between Dates 01-JUL-2026:00:00:00 and 30-AUG-2026:00:00:00										
	Group Observations 7, 8, 9 within 1 Days										

Proposal 12586 - Observation 10 - Pioneering Supernova Dust Studies with JWST: Detection of Dust Precursors in SN 2024ahv

Observation	Proposal 12586, Observation 10: NIRSPEC											Tue Jun 09 18:01:23 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 10:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	2024ahv	RA: 16 18 46.3190 (244.6929958d)				Proper Motion RA: 0					
			Dec: +07 24 44.77 (7.41244d)				Proper Motion Dec: 0					
			Equinox: J2000				Epoch of Position: 2000					
	Comments: Category=Star Description=[Supernovae] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	WATA	SUB2048	CLEAR	NRSRAPIDD6	3	1	1	14.452	227508.2	
Template	HFF Readout Mode				Slit				Subarray			
	false				S400A1				SUBS400A1			
Dithers	#	Primary Dither Positions							Sub-Pixel Pattern			
	1	3							NONE			
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235M/F170LP	S400A1	NRSRAPID	100	1	1	NONE	3	3	472.135	227508.4
	2	G395M/F290LP	S400A1	NRSRAPID	100	2	2	NONE	3	6	944.271	227508.6

Proposal 12586 - Observation 10 - Pioneering Supernova Dust Studies with JWST: Detection of Dust Precursors in SN 2024ahv	
Special Requirements	Between Dates 10-FEB-2027:00:00:00 and 20-APR-2027:00:00:00 Sequence Observations 10, 11, 12 within 1 Days

Proposal 12586 - Observation 11 - Pioneering Supernova Dust Studies with JWST: Detection of Dust Precursors in SN 2024ahv

Observation	Proposal 12586, Observation 11: MIRI/LRS										Tue Jun 09 18:01:23 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: MIRI Low Resolution Spectroscopy										
	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	(Visit 11:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
Acquisition	(1)	2024ahv	RA: 16 18 46.3190 (244.6929958d)			Proper Motion RA: 0					
			Dec: +07 24 44.77 (7.41244d)			Proper Motion Dec: 0					
Template			Equinox: J2000			Epoch of Position: 2000					
	Comments: Category=Star Description=[Supernovae] Extended=NO										
Dithers	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	SAME	F560W	FAST	10	1	1	27.75	227508.1		
Pointing Verification	Subarray										
	FULL										
Pointing Verification	Obtain Verification Image?										
	true										
Pointing Verification	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset	
	1	ALONG SLIT NOD									
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter	
	1	FASTR1	10	1	1	1	1	27.75		F560W	

Proposal 12586 - Observation 11 - Pioneering Supernova Dust Studies with JWST: Detection of Dust Precursors in SN 2024ahv

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	100	10	20	1	2	5600.031	227508.3
Special Requirements	Between Dates 10-FEB-2027:00:00:00 and 20-APR-2027:00:00:00								
	Sequence Observations 10, 11, 12 within 1 Days								

Proposal 12586 - Observation 12 - Pioneering Supernova Dust Studies with JWST: Detection of Dust Precursors in SN 2024ahv

Observation	Proposal 12586, Observation 12: MIRI/Imaging										Tue Jun 09 18:01:23 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 12:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	2024ahv	RA: 16 18 46.3190 (244.6929958d)			Proper Motion RA: 0					
			Dec: +07 24 44.77 (7.41244d)			Proper Motion Dec: 0					
			Equinox: J2000			Epoch of Position: 2000					
	Comments: Category=Star Description=[Supernovae] Extended=NO										
Template	Subarray										
	BRIGHTSKY										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				1	1	POINT SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F1280W	FASTR1	50	4	1	Dither 1	4	16	702.607	227508.7
	2	F1500W	FASTR1	50	4	1	Dither 1	4	16	702.607	227508.9
	3	F1800W	FASTR1	70	4	1	Dither 1	4	16	979.497	227508.10
	4	F2100W	FASTR1	70	8	1	Dither 1	4	32	1962.455	227508.11
	5	F2550W	FASTR1	50	15	1	Dither 1	4	60	2644.296	227508.12
Special Requirements	Between Dates 10-FEB-2027:00:00:00 and 20-APR-2027:00:00:00										
	Sequence Observations 10, 11, 12 within 1 Days										